

Notes: Purnanandam (RFS, 2011)

Originate-to-Distribute Model and the Subprime Mortgage Crisis

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1 Big picture

- **Question.** Did the originate-to-distribute (OTD) model weaken mortgage screening incentives and thereby contribute to the subprime mortgage crisis?
- **Central mechanism.** In a traditional originate-to-hold model, the bank bears the long-run credit risk of the borrower, so it has strong incentives to collect soft information and screen carefully. In an OTD model, the bank originates the loan but intends to sell it to outside investors. That separation between origination and ultimate risk-bearing can dilute screening incentives.
- **Main hypothesis.** Banks that were more aggressive users of the OTD model before the crisis originated mortgages with worse *unobserved* quality. When the secondary mortgage market froze in mid-2007, these banks were left holding more of the bad loans they had planned to sell, and their mortgage defaults rose more sharply.
- **Why the crisis is useful for identification.** During normal times, even a bank that plans to sell a loan may not be exposed to much warehousing risk because it can quickly move the loan off its balance sheet. Once the secondary market became illiquid, that option disappeared. The crisis therefore reveals the quality of loans that high-OTD banks had been originating earlier.
- **Empirical design.** The paper compares banks with different pre-crisis OTD intensity before and after the disruption in the mortgage resale market. It then strengthens the design with matched-sample tests using HMDA loan-level data and with heterogeneity tests based on bank capital and funding structure.
- **Headline findings.**
 - Banks with high pre-crisis OTD activity were less able to sell mortgages after 2007Q1.
 - Those same banks experienced significantly larger mortgage chargeoffs and non-performing mortgage ratios after the crisis.
 - The effects are stronger when the bank is measured to be *stuck* with OTD loans that it originated before the crisis but could not sell later.
 - The results remain strong after matching banks on geography, size, borrower income, loan-to-income ratios, and even average rate spreads charged to risky borrowers.
 - High-OTD banks exhibit tighter residual loan-spread distributions, which the paper interprets as evidence of weaker screening on soft information.
 - The OTD-default link is strongest for low-capital banks and for banks with less reliance on demand deposits.
- **What the paper rules out.** The paper argues that the results are not simply driven by observable borrower risk, geographic differences in house prices, or a lower cost of capital for high-OTD banks.
- **Economic interpretation.** The paper supports a classic agency story: when the originator does not expect to keep the loan, the incentive to acquire costly private information declines. In the run-up to the crisis, that incentive problem appears to have been economically important.

- **Broader contribution.** The paper ties the mortgage crisis not only to securitization itself, but to the incentive effects created by securitization. It also shows that a bank’s liability structure matters for how aggressively it uses OTD lending.
- **Bottom line.** The main message is that OTD lending was not just a funding technology. It changed underwriting incentives, and those distorted incentives materially worsened mortgage performance once market liquidity disappeared.

2 Data and measurement (key objects)

2.1 Main data sources

The paper combines two datasets.

Call reports. Quarterly FDIC call reports provide bank-level information on balance sheets, loan composition, mortgage chargeoffs, non-performing mortgages, capital, liquidity, deposits, and mortgage sales activity. Beginning in 2006Q3, banks also report:

- 1-4 family residential mortgages originated during the quarter with a purpose to resell, and
- 1-4 family residential mortgages actually sold during the quarter.

These items come from schedule RC-P and are available for banks with at least \$1 billion in assets, plus smaller banks with sufficiently large mortgage-sale activity.

HMDA data. The Home Mortgage Disclosure Act database provides loan-level information on borrower income, loan amount, geography, neighborhood characteristics, lien status, and, for higher-risk loans, the loan rate spread. HMDA is used mainly to match otherwise similar banks and to control for observable borrower risk.

2.2 Sample construction

The bank-quarter sample runs from 2006Q3 through 2008Q1, giving seven quarters total. The paper intersects the call report sample with banks covered in HMDA in 2006 and then imposes a balanced-panel requirement so that the same bank is observed throughout the pre-crisis and post-crisis quarters.

In the full bank-quarter panel, there are 5,397 observations for most regressions, which corresponds to roughly 771 banks observed over seven quarters. The sample tilts toward relatively large banks because OTD reporting is required only for banks active enough in the mortgage-sale market.

2.3 Key timing

The paper treats the period up to and including 2007Q1 as the **pre-disruption** period and quarters after 2007Q1 as the **post-disruption** period. The underlying event is the sudden loss of liquidity in the secondary mortgage market during mid-2007.

The post-crisis indicator is therefore:

$$\text{after}_t = \begin{cases} 0 & \text{for } t \leq 2007Q1, \\ 1 & \text{for } t \geq 2007Q2. \end{cases} \quad (1)$$

2.4 Main OTD variable

The central bank-level exposure measure is **preotd**, defined as the bank's average pre-crisis intensity of mortgage origination for resale:

$$\text{preotd}_i = \frac{1}{3} \sum_{q \in \{2006Q3, 2006Q4, 2007Q1\}} \frac{\text{Mortgage originations for resale}_{iq}}{\text{Beginning-of-quarter mortgage loans}_{iq}}. \quad (1)$$

Interpretation. This is not a securitization dummy. It is a *continuous measure* of how strongly bank i relied on the OTD business model before the market broke down.

A few descriptive statistics are useful:

- Mean preotd = 0.23.
- Median preotd = 0.05.
- Maximum preotd = 3.06.

So the OTD distribution is very skewed: many banks have little activity, but a subset relies heavily on the model.

2.5 Mortgage sale intensity

The paper also measures the fraction of mortgages the bank actually sells in a given quarter:

$$\text{sold}_{it} = \frac{\text{Mortgage loans sold during quarter } t}{\text{Beginning-of-quarter mortgage loans}}. \quad (2)$$

This variable is important because the identification strategy first shows that high-OTD banks experienced a sharper decline in actual loan sales after the market disruption.

2.6 Mortgage performance outcomes

The paper uses two main measures of mortgage quality.

Mortgage chargeoff rate.

$$\text{ChargeoffRate}_{it} = \frac{\text{Net chargeoffs on 1-4 family mortgages}_{it}}{\text{Beginning-of-quarter mortgage loans}_{it}}. \quad (3)$$

Mortgage non-performing asset rate (mortNPA).

$$\text{mortNPA}_{it} = \frac{\text{Non-performing 1-4 family mortgages}_{it}}{\text{Mortgage loans}_{it}}. \quad (4)$$

The paper prefers mortgage NPA as the cleaner default measure because chargeoffs may be influenced by accounting discretion. Still, the chargeoff regressions tell the same story.

Key summary statistics from Table 1:

- Mean mortgage NPA ratio = 2.03%.
- Median mortgage NPA ratio = 1.35%.
- Mean mortgage chargeoff ratio = 0.04%.

2.7 The “stuck” measure

The paper refines the exposure measure with a proxy for loans that were originated with the intent to distribute but could not be distributed after the crisis:

$$\text{stuck}_i = \overline{\text{OTD originations}}_{i, 2006Q3:2007Q1} - \overline{\text{Loans sold}}_{i, 2007Q2:2008Q1}, \quad (5)$$

where both components are scaled by pre-crisis mortgage assets.

Interpretation. A high value of stuck means the bank built up a large pipeline of pre-crisis OTD loans and then failed to unload enough of them once market liquidity disappeared.

2.8 Controls and heterogeneity variables

The bank-level controls include:

- **premortgage:** average mortgage loans to total assets before the crisis,
- **logta:** log total assets,
- **cil/ta:** commercial and industrial loans to total assets,
- **liquid:** liquid assets to total assets,
- **absgap:** absolute one-year maturity gap scaled by total assets.

For heterogeneity, the paper focuses on:

- **cap:** average tier-one capital ratio in the pre-crisis quarters,
- **dd:** average demand deposits to total deposits in the pre-crisis quarters,
- HMDA-based bank-level risk summaries such as average loan-to-income ratio (li), share of high-rate loans, and share of no-income-documentation loans.

2.9 Why the descriptive evidence already matters

Several raw facts are already suggestive.

- OTD originations average just under 30% of mortgage portfolios in 2006Q3-2006Q4 and then fall to about 20% after the disruption.
- Actual loan sales decline sharply after 2007Q1.

- Mortgage chargeoffs increase about fourfold from 2007Q1 to 2007Q4.
- In Figure 4, high-OTD and low-OTD banks have similar pre-crisis chargeoffs but sharply diverging post-crisis outcomes.

These patterns motivate the formal difference-in-differences tests.

3 Models and empirical specifications (all equations + interpretation)

3.1 First step: did high-OTD banks lose the ability to sell loans?

The paper first estimates whether banks with high pre-crisis OTD exposure experienced a sharper decline in mortgage sales after the disruption:

$$\text{sold}_{it} = \beta_0 + \beta_1 \text{after}_t + \beta_2 \text{preotd}_i + \beta_3 (\text{after}_t \times \text{preotd}_i) + \beta_4 (\text{after}_t \times \text{premortgage}_i) + \Gamma' X_{it} + \varepsilon_{it}. \quad (6)$$

In fixed-effect versions, the time-invariant levels of `preotd` and `premortgage` are absorbed by bank fixed effects.

Interpretation of β_3 . A negative coefficient means that banks with higher pre-crisis OTD exposure sold a smaller fraction of mortgages after the disruption. That is the warehousing-risk step of the argument.

3.2 Baseline default regression

The core bank fixed-effect specification is:

$$\text{default}_{it} = \mu_i + \beta_1 \text{after}_t + \beta_2 (\text{after}_t \times \text{preotd}_i) + \beta_3 (\text{after}_t \times \text{premortgage}_i) + \Gamma' X_{it} + \varepsilon_{it}. \quad (7)$$

Here:

- μ_i are bank fixed effects,
- `default` is either the mortgage chargeoff rate or the mortgage NPA ratio,
- `after` captures the common jump in defaults after the crisis,
- `after × preotd` is the coefficient of interest,
- and `after × premortgage` makes sure the result is not just a mortgage-specialization effect.

Interpretation of β_2 . A positive β_2 means that high-OTD banks saw a larger increase in mortgage defaults after the crisis than low-OTD banks, conditional on bank fixed effects and controls.

3.3 Mechanism regression using stuck loans

To tie the result more directly to illiquidity in the resale market, the paper re-estimates the default equation with stuck replacing preotd:

$$\text{default}_{it} = \mu_i + \beta_1 \text{after}_t + \beta_2 (\text{after}_t \times \text{stuck}_i) + \beta_3 (\text{after}_t \times \text{premortgage}_i) + \Gamma' X_{it} + \varepsilon_{it}. \quad (8)$$

Interpretation. If the story is really about loans that banks intended to distribute but were forced to keep, then $\text{after} \times \text{stuck}$ should predict post-crisis defaults at least as well as $\text{after} \times \text{preotd}$. That is exactly what the paper finds.

3.4 Matched-sample specifications

The paper then estimates versions of Equation (7) and Equation (8) on matched samples of high- and low-OTD banks. The matching is not random, but it greatly narrows observable differences.

Main matching dimensions include:

- primary state of lending,
- bank size group,
- borrower income,
- loan-to-income ratio,
- in some tests, average rate spread on risky loans,
- in some unreported tests, subprime share and other risk measures.

Interpretation. The matched designs try to compare banks that faced very similar borrowers and locations before the crisis, so that the remaining post-crisis difference is more plausibly about screening incentives rather than borrower composition.

3.5 Regression with observed borrower-risk interactions

As a complement to exact matching, the paper also estimates a specification that interacts the post-crisis dummy with bank-level summaries of borrower risk:

$$\text{default}_{it} = \mu_i + \beta_1 \text{after}_t + \beta_2 (\text{after}_t \times \text{preotd}_i) + \sum_{m=1}^M \theta_m (\text{after}_t \times \text{risk}_{im}) + \Gamma' X_{it} + \varepsilon_{it}. \quad (9)$$

The omitted detailed results still support the core OTD effect.

3.6 Pricing equation for the screening test

To distinguish “risky but properly priced” from “risky because screening was weak,” the paper studies the dispersion of loan-rate residuals from:

$$\text{rate}_{ib} = \alpha + \Lambda' X_{ib} + u_{ib}, \quad (10)$$

where X_{ib} includes borrower, loan, neighborhood, and bank observables for loan i issued by bank b . The logic is simple:

- a bank that screens carefully should charge more differentiated rates across borrowers with different soft information,
- a bank that screens weakly should have less dispersion in residual loan pricing.

Interpretation. Smaller residual spread dispersion for high-OTD banks is evidence consistent with weaker screening on unobserved borrower quality.

3.7 Triple-difference model: bank capital

To understand which banks produced the worst OTD loans, the paper estimates:

$$\text{default}_{it} = \mu_i + \beta_1 \text{after}_t + \beta_2 (\text{after}_t \times \text{preotd}_i) + \beta_3 (\text{after}_t \times \text{cap}_i) + \beta_4 (\text{after}_t \times \text{preotd}_i \times \text{cap}_i) + \Gamma' X_{it} + \varepsilon_{it}. \quad (11)$$

Interpretation of β_4 . If $\beta_4 < 0$, then the default effect of OTD lending is weaker for well-capitalized banks. That means the problematic OTD lending is concentrated among banks with low capital.

3.8 Triple-difference model: demand deposits

The paper runs an analogous specification for demand-deposit funding:

$$\text{default}_{it} = \mu_i + \beta_1 \text{after}_t + \beta_2 (\text{after}_t \times \text{preotd}_i) + \beta_3 (\text{after}_t \times \text{dd}_i) + \beta_4 (\text{after}_t \times \text{preotd}_i \times \text{dd}_i) + \Gamma' X_{it} + \varepsilon_{it}. \quad (12)$$

Interpretation of β_4 . If $\beta_4 < 0$, then greater reliance on demand deposits attenuates the OTD-default link. The paper interprets this as evidence that fragile demandable funding can act as a disciplining device.

3.9 What all the equations are trying to isolate

Across these specifications, the core empirical question is always the same: *holding fixed broad bank characteristics and observable borrower risk, do banks that relied more on originate-to-distribute lending before the crisis suffer worse mortgage performance after the crisis?*

The paper's answer is yes.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no external instrument

This paper does *not* use a conventional instrumental-variable design. Its identification comes from a quasi-experimental timing argument: the secondary mortgage market suddenly became illiquid

in mid-2007, which exposed the quality of loans originated earlier by banks with different OTD intensity.

4.2 Why the shock is informative

If a bank originated weakly screened loans but could sell them quickly in normal times, the bank would not obviously suffer later defaults on its own balance sheet. Once the market froze, that exit option disappeared. The crisis therefore turned earlier OTD behavior into observable balance-sheet risk.

4.3 Difference-in-differences logic

The main design compares:

- banks with high versus low pre-crisis OTD participation,
- before versus after the market disruption.

The identifying coefficient is the interaction between post-crisis time and pre-crisis OTD exposure. Bank fixed effects absorb time-invariant differences across banks. The post dummy captures common deterioration in mortgage markets. The interaction isolates whether high-OTD banks deteriorated more.

4.4 Graphical support for the design

Figure 4 is important because it shows:

- similar chargeoff levels for high- and low-OTD banks before the crisis,
- parallel behavior before 2007Q1,
- and a sharp divergence only after the disruption.

That pattern is exactly what the difference-in-differences design requires.

4.5 Separating OTD from plain mortgage specialization

A natural concern is that banks with high OTD were simply big mortgage lenders, and mortgage lenders in general were hit harder by the crisis. The paper addresses this by controlling for premortgage and interacting it with the post-crisis indicator. That forces the OTD coefficient to capture more than just high mortgage exposure.

4.6 Mechanism sharpening with stuck loans

The stuck variable makes the design more credible because it directly targets the subset of loans that high-OTD banks originated before the crisis but failed to sell later. The stronger results for after $\times$ stuck support the warehousing-risk mechanism rather than a vague cross-sectional correlation.

4.7 Matched samples to handle borrower observables

A second major concern is selection on borrower observables. Maybe high-OTD banks simply lent to riskier borrowers. The matched-sample exercises sharply reduce this concern by comparing banks that look similar in:

- state-level geographic exposure,
- size,
- average borrower income,
- average loan-to-income ratio,
- and, in stricter designs, average risky-loan spreads.

The fact that the post-crisis OTD effect survives in these matched samples is a major credibility point.

4.8 Why matching on interest rates is especially important

Matching on interest rates goes after a subtle alternative story: maybe high-OTD banks did originate riskier loans, but they understood those risks and priced them correctly. If that were true, conditioning on loan spreads should absorb the difference. But the OTD-default effect remains strong even when matched banks charge similar rates. That is hard to reconcile with “they screened fine and priced correctly.”

4.9 Cost of capital versus screening

The paper also considers the Pennacchi-style argument that OTD lowers cost of capital and therefore allows banks to fund riskier but still positive-NPV borrowers. To challenge that explanation, it compares *small* high-OTD banks with *large* low-OTD banks. Since large banks are unlikely to have a higher cost of capital than the small banks, the persistence of the OTD effect in this sample weakens the pure cost-of-capital story.

4.10 Soft-information evidence from loan-spread dispersion

The shrinkage in residual loan-spread dispersion is not the main causal design, but it adds important supporting evidence. If high-OTD banks gathered and used soft information as carefully as low-OTD banks, they should price otherwise similar risky borrowers more differentially. They do not. The narrower spread distribution is consistent with weaker soft-information screening.

4.11 Remaining limitations

The paper is careful, but the design is not perfect.

- The crisis is not randomly assigned.
- The main OTD measure begins only in 2006Q3, so there is limited pre-period data.

- The stuck measure is a proxy, not a loan-level match between originations and later unsold loans.
- The main matching is at the state level rather than at finer geographic units, though the paper reports similar robustness for narrower matching in unreported tests.
- The paper is mostly cross-sectional, so it cannot fully settle broader macro explanations for the mortgage crisis.

Still, taken together, the fixed effects, matched samples, mechanism tests, and heterogeneity results make the paper's underwriting-incentives interpretation highly persuasive.

5 Tables and figures

5.1 Figure 1, Figure 2, and Figure 3: what happened around the crisis?

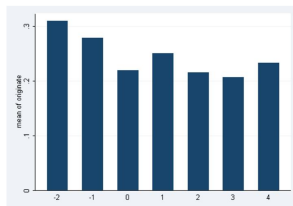


Figure 1
Mortgage originated for distribution over time
The figure plots the ratio of OTD loans to total mortgages on a quarterly basis. We plot the average value of this ratio across all banks with available information in the sample. Quarter zero corresponds to the quarter ending on March 31, 2007.

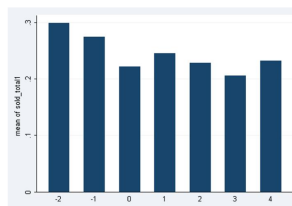


Figure 2
Mortgage sold over time
The figure plots the extent of loans sold as a fraction of mortgages outstanding as of the beginning of the quarter. We plot the average value of this ratio across all banks with available information in the sample. Quarter zero corresponds to the quarter ending on March 31, 2007.

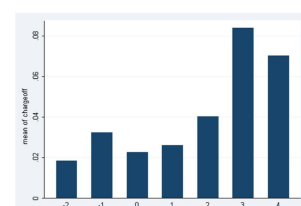


Figure 3
Mortgage chargeoff over time
The figure plots the average net chargeoff as a percentage of mortgage outstanding on a quarterly basis. Quarter zero corresponds to the quarter ending on March 31, 2007.

Read:

- Figure 1 shows that mortgage originations for resale were high before the disruption, averaging just under 30% of mortgage portfolios in 2006Q3 and 2006Q4, and then dropping to around 20% afterward.
- Figure 2 shows a clear decline in actual mortgage sales after 2007Q1.
- Figure 3 shows that mortgage chargeoffs rose steadily and increased roughly fourfold from 2007Q1 to 2007Q4.

Economic message. The basic environment changed sharply in mid-2007: banks wanted to originate fewer OTD loans, sold fewer loans, and the loans they retained performed much worse.

5.2 Table 1 and Figure 4: who is in the sample, and what do the raw trends say?

Read:

- Table 1 shows that the average bank in the sample has \$5.92 billion in assets, a mortgage-to-assets ratio of 0.17, a tier-one capital ratio of 0.11, and a demand-deposit share of 0.09.
- The mean mortgage NPA ratio is 2.03%, with a median of 1.35%.

Table 1
Summary statistics

variable	N	mean	p50	min	max
ta	5397.00	5.92	1.05	0.06	168.65
mortgage/ta	5397.00	0.17	0.15	0.01	0.49
cil/ta	5397.00	0.11	0.10	0.00	0.39
td/ta	5397.00	0.78	0.80	0.44	0.92
dd/d	5397.00	0.09	0.08	0.01	0.33
leverage	6636.00	0.90	0.91	0.77	0.94
nii/ta	5397.00	0.89	0.87	0.32	1.51
chargeoff(%)	5397.00	0.04	0.00	-0.07	0.79
npa/ta(%)	5397.00	0.73	0.44	0.00	5.40
mortnpa(%)	5397.00	2.03	1.35	0.00	13.86
tier1cap	5397.00	0.11	0.10	0.07	0.29
liquid	5397.00	0.15	0.12	0.02	0.50
absgap	5397.00	0.14	0.11	0.00	0.51
preotd	771.00	0.23	0.05	0.00	3.06

This table provides the summary statistics of key variables used in the study. All variables are computed using call report data for seven quarters starting from 2006Q3 and ending in 2008Q1. We provide the number of observations (N), mean, median, minimum, and maximum values for each variable. ta is total assets in billions of dollars; $mortgage/ta$ is the ratio of 1–4 family residential mortgages outstanding to total assets; cil/ta is the ratio of commercial and industrial loans to total assets; td/ta is the ratio of total deposits to total assets; dd/d is the ratio of demand deposits to total deposits; nii/ta is the ratio of net interest income to total assets; $chargeoff$ measures the chargeoff on mortgage portfolio (net of recoveries) as a percentage of mortgage assets; npa/ta is the ratio of non-performing assets to total assets; $mortnpa$ is the ratio of non-performing mortgages to total mortgages; $tier1cap$ measures the ratio of tier-one capital to risk-adjusted assets; $liquid$ is the bank's liquid assets to total assets ratio; $absgap$ is the absolute value of one-year maturity gap as a fraction of total assets. $preotd$ measures the originate-to-distribute loans, i.e., mortgages originated with a purpose to sell, as a fraction of total mortgages. This variable is constructed at the bank level based on its average quarterly values during 2006Q3, 2006Q4, and 2007Q1.

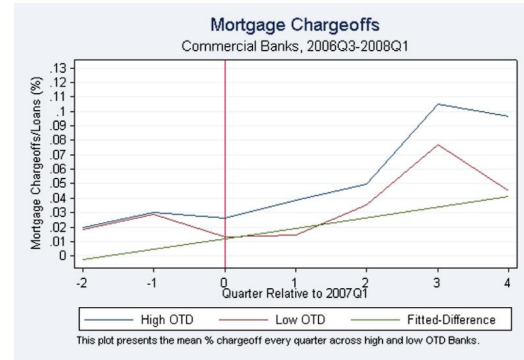


Figure 4
Mortgage chargeoff and OTD participation
The figure plots the average net chargeoff (as a percentage of mortgage outstanding) on the bank's mortgage portfolio across two groups of banks sorted on the basis of their participation in the OTD market prior to March 31, 2007.

- Mean $preotd$ is 0.23, but the median is only 0.05, indicating strong skewness in OTD exposure.
- Figure 4 sorts banks by pre-crisis OTD participation. Both high- and low-OTD banks start with similar chargeoff rates before the crisis, but the high-OTD group experiences a much steeper post-crisis increase.

Economic message. The descriptive evidence already looks like a classic difference-in-differences pattern: similar pre-trends, then a sharp divergence once the mortgage market breaks.

5.3 Table 2: high-OTD banks became less able to sell mortgages

Read:

- In the bank fixed-effect specification, the coefficient on $after*preotd$ is about -0.204 in Model 2 and -0.212 in Model 3.
- The negative sign means banks with more pre-crisis OTD activity suffered a larger decline in mortgage sales after the disruption.
- The paper interprets the estimate as economically large: a one-standard-deviation increase in $preotd$ implies about a 10% decline in selling intensity after the crisis.

Economic message. This table validates the warehousing-risk channel. The crisis did not hit all banks' resale capacity equally; it hit high-OTD banks especially hard.

5.4 Table 3: the baseline default result

Read:

- For all banks, $after*preotd$ is 0.0420 in the chargeoff regression and 0.4439 in the mortgage-NPA regression.
- Excluding banks with more than \$10 billion in assets barely changes the result: the NPA coefficient remains about 0.4015.

Table 2
Intensity of mortgages sold

	Model 1		Model 2		Model 3	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
<i>preotd</i>	0.9591	(54.64)				
<i>premortgage</i>	0.0403	(0.85)				
<i>after</i>	0.0273	(1.95)	0.0182	(1.23)	0.0205	(1.24)
<i>after*preotd</i>	-0.1889	(-3.34)	-0.2037	(-3.74)	-0.2120	(-3.86)
<i>after*premortgage</i>	0.0163	(0.21)	0.0235	(0.29)	0.0428	(0.49)
<i>logta</i>	-0.0031	(-0.54)	0.1475	(2.88)	0.1575	(2.44)
<i>cil/ta</i>	-0.0248	(-0.22)	-0.8606	(-2.74)	-0.7744	(-2.40)
<i>liquid</i>	0.0339	(0.48)	-0.0292	(-0.21)	0.0570	(0.38)
<i>absgap</i>	-0.0320	(-0.55)	0.2866	(2.79)	0.3171	(2.82)
R^2	0.8156		0.9039		0.9054	
N	4476		4476		4100	
State dummies	Yes		No		No	
Bank fixed effect	No		Yes		Yes	
Exclude large banks	No		No		Yes	

This table provides the regression results of the following model:

$$sold_{it} = \beta_0 + \beta_1 after_{it} + \beta_2 preotd_{it} + \beta_3 after_{it} * preotd_{it} + \sum_{k=1}^{k=K} \beta X + \epsilon_{it}.$$

The dependent variable, $sold_{it}$, measures bank i 's mortgage sale as a fraction of its total mortgage loans at the beginning of quarter t . $after_{it}$ is a dummy variable that is set to zero for quarters before and including 2007Q1, and one after that. $preotd_{it}$ is the average value of OTD mortgages to total mortgages during quarters 2006Q3, 2006Q4, and 2007Q1. X stands for a set of control variables. Model 1 is estimated using the OLS method. Models 2 and 3 are estimated with bank fixed effects. Model 3 excludes banks with more than \$10 billion in assets. These models omit $preotd$ and $premortgage$ as right-hand-side variables since they remain constant across all seven quarters for a given bank. $premortgage$ is the average ratio of mortgage assets to total assets for 2006Q3, 2006Q4, and 2007Q1. $logta$ measures the log of total assets; cil/ta is the ratio of commercial and industrial loans to total assets; $liquid$ is the bank's liquid assets to total asset ratio; $absgap$ is the absolute value of one-year maturity gap as a fraction of total assets. Adjusted R -squared and number of observations are provided in the bottom rows. All standard errors are clustered at the bank level.

Table 3
Mortgage defaults

Dependent Var:	All Banks				Excludes Large Banks			
	Model 1 Chargeoffs		Model 2 NPA		Model 3 Chargeoffs		Model 4 NPA	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
<i>after</i>	0.0116	(1.91)	0.3411	(3.03)	0.0134	(2.03)	0.3076	(2.52)
<i>after*preotd</i>	0.0420	(2.76)	0.4439	(2.44)	0.0428	(2.76)	0.4015	(2.21)
<i>after*premortgage</i>	0.0060	(0.21)	0.6600	(1.22)	-0.0062	(-0.20)	0.4261	(0.82)
<i>logta</i>	0.0925	(4.15)	0.2266	(0.51)	0.0776	(2.67)	0.6896	(1.42)
<i>cil/ta</i>	0.2010	(1.65)	2.6103	(1.37)	0.1662	(1.32)	2.2626	(1.13)
<i>liquid</i>	0.0745	(1.24)	1.3732	(0.90)	0.1089	(1.64)	-0.1540	(-0.14)
<i>absgap</i>	-0.0672	(-1.59)	-3.2639	(-3.85)	-0.0742	(-1.64)	-3.1248	(-3.66)
R^2	0.3805		0.7297		0.3621		0.7135	
N	5397		5397		4977		4977	

This table provides the regression results of the following fixed-effects model:

$$default_{it} = \mu_i + \beta_1 after_{it} + \beta_2 after_{it} * preotd_{it} + \sum_{k=1}^{k=K} \beta X + \epsilon_{it}.$$

The dependent variable, $default_{it}$, is measured by either the mortgage chargeoffs or the non-performing mortgages (scaled by the outstanding mortgage loans) of bank i during quarter t . $after_{it}$ is a dummy variable that is set to zero for quarters before and including 2007Q1, and one after that. $preotd_{it}$ is the average value of OTD mortgages to total mortgages during quarters 2006Q3, 2006Q4, and 2007Q1. μ_i denotes bank fixed effects; X stands for a set of control variables. $premortgage$ is the average ratio of mortgage assets to total assets for 2006Q3, 2006Q4, and 2007Q1. $logta$ measures the log of total assets; cil/ta is the ratio of commercial and industrial loans to total assets; $liquid$ is the bank's liquid assets to total assets ratio; $absgap$ is the absolute value of one-year maturity gap as a fraction of total assets. Adjusted R -squared and number of observations are provided in the bottom rows. All standard errors are clustered at the bank level.

- The paper states that a one-standard-deviation increase in *preotd* raises mortgage default rates by about 11% relative to the unconditional sample mean.

Economic message. Banks that relied more on the OTD model before the crisis experienced meaningfully larger mortgage losses afterward, even after controlling for fixed bank characteristics and the overall time trend.

5.5 Table 4: the effect is strongest for banks stuck with unsold OTD loans

Read:

- The coefficient on *after*stuck* is 0.0922 for chargeoffs and 1.4342 for mortgage NPAs in the full sample.
- The corresponding coefficients remain strong when large banks are excluded.
- The paper notes that in horse-race tests, the effect of OTD lending on mortgage defaults mainly comes from the variation in *stuck*.

Economic message. The bad performance of high-OTD banks is not a generic cross-sectional fact. It is tightly linked to the loans they originated before the crisis and then failed to unload.

5.6 Figure 5 and Table 5: matched-sample evidence based on borrower observables

Read:

- Figure 5 shows that matched high- and low-OTD banks have nearly identical distributions of borrower income and loan-to-income ratios.

Table 4

Mortgage default and inability to sell

Dependent Var:	All Banks				Excludes Large Banks			
	Model 1 Chargeoffs		Model 2 NPA		Model 3 Chargeoffs		Model 4 NPA	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat

This table provides regression results for the following fixed-effect model:

$$default_{it} = \alpha_i + \beta_1 after_{it} + \beta_2 after*preotd_{it} + \beta_3 after*stuck_{it} + \sum_{k=1}^{K-1} \beta_{k+1} EX_k + \varepsilon_{it}$$

The dependent variable, $default_{it}$, is measured by either the mortgage chargeoffs or the non-performing mortgages of bank i during quarter t . $after_{it}$ is a dummy variable that is set to zero for quarters before and including 2007Q1, and one after that. $stuck_{it}$ measures the difference between loans originated before 2007Q1 and loans sold after that quarter. $preotd_{it}$ denotes bank fixed effects. K stands for a set of control variables. $premortgage$ is the average ratio of mortgage assets to total assets for 2006Q3, 2006Q4, and 2007Q1. $logta$ measures the log of total assets; $clira$ is the ratio of commercial and industrial loans to total assets; $liquid$ is the bank's liquid assets to total assets ratio; $abgap$ is the absolute value of one-year maturity gap as a fraction of total assets. Adjusted R -squared and number of observations are provided in the bottom rows. All standard errors are clustered at the bank level.

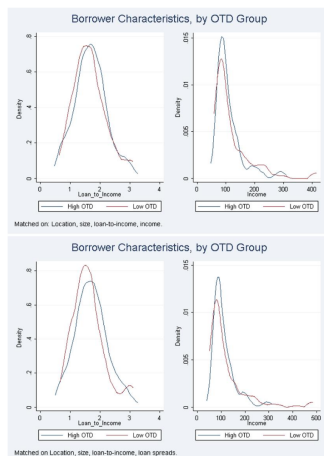


Figure 5
Distribution of key characteristics of high- and low-OTD banks after matching
The plots give the kernel density functions of the key characteristics of the high- and low-OTD banks after matching. More details on the matching are provided in the article. The first plot is for the loan-to-income ratio; the second plot is for the borrowers' annual income.

Table 5

Matched sample analysis: Base case

Dependent Var:	Model 1 NPA		Model 2 NPA		Model 3 NPA		Model 4 NPA	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat

This table reports the estimation results of fixed-effect regressions on a matched sample of high- and low-OTD banks. Banks are matched on geographical location of their mortgage portfolios, the borrowers' loan-to-income ratio, the borrowers' annual income, and the bank's size. The dependent variable is the non-performing mortgage loans of banks in a given quarter. The definitions of variables and details of the model estimation are provided in the article. Adjusted R -squared and number of observations are provided in the bottom rows. All standard errors are clustered at the bank level.

- In Table 5, the coefficient on $after*preotd$ is about 0.90, and the coefficient on $after*stuck$ is about 2.36 to 2.39.
- The paper translates the magnitude into about 0.45 percentage points higher default for banks with one-standard-deviation-higher OTD exposure, or roughly 32% of the sample median default rate.

Economic message. Once the paper holds geography and observable borrower quality roughly constant, the OTD effect does not disappear. If anything, it becomes stronger.

5.7 Table 6: matched-sample evidence based on interest rates

Read:

- When banks are additionally matched on the average spread charged to risky borrowers, $after*preotd$ remains about 0.656 to 0.688.
- $after*stuck$ remains around 1.74 to 1.82.
- The paper reports that a one-standard-deviation increase in $preotd$ raises default by about 0.38 percentage points, or 26% of the matched-sample median default rate.

Economic message. High-OTD banks did not just lend to riskier borrowers and price that risk properly. Even among banks charging similar risky-loan spreads in similar places, post-crisis performance is worse for high-OTD lenders.

5.8 Table 7: small high-OTD banks still underperform large low-OTD banks

Read:

- In the small-to-big matched sample, $after*preotd$ is about 0.84 to 0.87.
- $after*stuck$ is around 2.04 to 2.21.

Table 6
Matched sample analysis: Average loan spread charged

Dependent Var:	Model 1 NPA		Model 2 NPA		Model 3 NPA		Model 4 NPA	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
<i>after</i>	0.5060	(5.58)	0.4911	(5.79)	0.2791	(1.53)	0.2505	(1.40)
<i>after*preotd</i>	0.6557	(2.28)			0.6880	(2.44)		
<i>after*stuck</i>			1.7369	(3.13)			1.8194	(3.33)
<i>after*premortgage</i>					0.4686	(0.64)	0.4564	(0.60)
<i>logta</i>					0.4861	(0.86)	0.6906	(1.22)
<i>cil/ta</i>					0.3309	(0.12)	0.3722	(0.13)
<i>absgap</i>					-4.5193	(-3.31)	-4.5168	(-3.29)
<i>liquid</i>					-2.1793	(-1.35)	-2.3290	(-1.46)
R^2	0.7049		0.7098		0.7118		0.7170	
N	2205		2205		2205		2205	

This table reports the estimation results of fixed-effect regressions on a matched sample of high- and low-OTD banks. Banks are matched on geographical location of their mortgage portfolios, bank size, the borrowers' loan-to-income ratio, and the average rate spread on the subprime loan portfolio of the bank. The dependent variable is the non-performing mortgage loans of the banks in a given quarter. The definition of variables and details of the model estimation are provided in the article. Adjusted R -squared and number of observations are provided in the bottom rows. All standard errors are clustered at the bank level.

Table 7
Matched sample analysis: Small to big

Dependent Var:	Model 1 NPA		Model 2 NPA		Model 3 NPA		Model 4 NPA	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
<i>after</i>	0.4577	(4.18)	0.4377	(4.36)	-0.0232	(-0.11)	-0.0922	(-0.44)
<i>after*preotd</i>	0.8384	(2.33)			0.8665	(2.38)		
<i>after*stuck</i>			2.0376	(2.60)			2.2099	(2.81)
<i>after*premortgage</i>					1.4795	(1.43)	1.4727	(1.40)
<i>logta</i>					2.3183	(1.77)	2.7952	(2.28)
<i>cil/ta</i>					4.3085	(0.80)	3.2424	(0.59)
<i>absgap</i>					-4.7469	(-2.25)	-4.6843	(-2.29)
<i>liquid</i>					3.4007	(0.74)	3.1728	(0.71)
R^2	0.6966		0.7046		0.7117		0.7230	
N	1148		1148		1148		1148	

This table reports the estimation results of fixed-effect regressions on a matched sample of high- and low-OTD banks. We match small banks with large OTD lending with large banks with little to no OTD lending. The dependent variable is the non-performing mortgage loans of the banks in a given quarter. The definitions of variables and details of the model estimation are provided in the article. Adjusted R -squared and number of observations are provided in the bottom rows. All standard errors are clustered at the bank level.

- These results hold even though the high-OTD banks are much smaller than the low-OTD comparison banks.

Economic message. The results are hard to explain purely by saying that OTD lowers funding costs and therefore attracts riskier but efficiently priced borrowers. Even small high-OTD banks underperform much larger low-OTD banks.

5.9 Table 8: loan-spread dispersion is tighter for high-OTD banks

Read:

- In the full sample, the residual standard deviation of loan spreads is 0.2236 for high-OTD banks versus 0.2621 for low-OTD banks.
- In the matched sample, the standard deviation is 0.2056 versus 0.2627.
- The interquartile spread and the 90-10 spread are also uniformly smaller for high-OTD banks.
- Bartlett and Levene tests reject equality of variance with p-values of 0.0001.

Economic message. High-OTD banks appear to price risky borrowers less discriminately after observable controls are removed. That is consistent with weaker collection and use of soft information.

5.10 Table 9: the role of bank capital

Read:

- The coefficient on *after*preotd* remains positive at around 1.01 to 1.15.
- The triple-interaction coefficient *after*preotd*cap* is about -5.20 to -5.50 and statistically significant.
- The paper interprets the magnitude as implying that a one-standard-deviation decrease in capital raises defaults by about 0.18 percentage points, around 13% of the sample median default rate.

Economic message. The bad OTD lending is disproportionately concentrated among poorly capitalized banks. That pattern supports a risk-shifting and weak-screening interpretation rather than a benign capital-saving story.

Table 8
Shrinkage in loan spread

	Panel A: All Banks			Panel B: Matched Sample		
	High OTD	Low OTD	Shrinkage	High OTD	Low OTD	Shrinkage
Standard Deviation	0.2236	0.2621	0.0385	0.2056	0.2627	0.0571
P75-P25	0.3144	0.3559	0.0415	0.2731	0.3669	0.0938
P90-P10	0.5697	0.6767	0.1070	0.5172	0.6883	0.1711
Bartlett's p -value			0.0001			0.0001
Levene's p -value			0.0001			0.0001

This table provides the dispersion in loan spread across high- (above-median) and low- (below-median) OTD banks. Panel A is for all banks, and Panel B for the matched sample. We provide three measures of dispersion in log loan spreads: standard deviation, the difference between the 75th and 25th percentiles, and the difference between the 90th and 10th percentiles. Shrinkage measures the difference in dispersion across the high- and low-OTD banks. Bartlett's and Levene's p -values are for the null hypothesis that the variance of loan spreads for the high-OTD group equals the variance of loan spreads for the low-OTD group.

Table 9
The effect of bank capital

Dependent Var:	All Banks				Excludes Large Banks			
	Model 1 NPA		Model 2 NPA		Model 3 NPA		Model 4 NPA	
	Estimate	t -stat	Estimate	t -stat	Estimate	t -stat	Estimate	t -stat
<i>after</i>	0.2390	(1.06)	-0.6064	(-2.16)	0.1240	(0.51)	-0.7314	(-2.43)
<i>after*cap</i>	0.8111	(0.45)	0.9272	(0.55)	1.4960	(0.76)	1.8426	(1.01)
<i>after*preotd*cap</i>	-5.4985	(-2.05)	-5.2733	(-2.01)	-5.1978	(-1.89)	-5.2544	(-1.93)
<i>after*preotd</i>	1.1495	(2.49)	1.0360	(2.31)	1.0716	(2.25)	1.0056	(2.16)
<i>after*premortgage</i>	0.6313	(1.17)	0.8177	(1.56)	0.4066	(0.78)	0.6109	(1.16)
<i>logta</i>	0.2335	(0.53)	0.3191	(0.75)	0.7031	(1.45)	0.7281	(1.48)
<i>cilita</i>	2.5588	(1.34)	2.8332	(1.45)	2.2419	(1.12)	2.5009	(1.22)
<i>liquid</i>	1.3795	(0.91)	1.3492	(0.95)	-0.1292	(-0.12)	-0.1240	(-0.11)
<i>absgap</i>	-3.2334	(-3.81)	-3.0484	(-3.69)	-3.1012	(-3.61)	-3.0050	(-3.51)
<i>after*li</i>			0.2714	(2.64)			0.2827	(2.63)
<i>after*highrate</i>			1.9064	(3.85)			1.6838	(3.67)
<i>after*noincome</i>			0.6671	(2.42)			0.5810	(1.87)
R^2	0.7303		0.7340		0.7140		0.7160	
N	5397		5327		4977		4907	

This table provides the regression results of the following fixed-effect model:

$$\text{default}_{it} = \mu_i + \beta_1 \text{after}_t + \beta_2 \text{after}_t * \text{preotd}_t + \beta_2 \text{after}_t * \text{cap}_t + \beta_3 \text{after}_t * \text{preotd}_t * \text{cap}_t + \sum_{k=1}^{k=K} \beta_k X + \epsilon_{it}.$$

The dependent variable, default_{it} , is measured as the ratio of non-performing mortgages to the outstanding mortgage loans of bank i during quarter t . after_t is a dummy variable that is set to zero for quarters before and including 2007Q1, and one after that. preotd_t is the average value of OTD mortgages to total mortgages during quarters 2006Q3, 2006Q4, and 2007Q1; cap_t is bank i 's average tier-one capital ratio during quarters 2006Q3, 2006Q4, and 2007Q1; μ_i denotes bank fixed effects; X stands for a set of control variables. premortgage is the average ratio of mortgage assets to total assets for 2006Q3, 2006Q4, and 2007Q1. logta measures the log of total assets; cilita is the ratio of commercial and industrial loans to total assets; liquid is the bank's liquid assets to total assets ratio; absgap is the absolute value of one-year maturity gap as a fraction of total assets. li measures the average loan-to-income ratio of all loans issued by the bank in 2006. highrate measures the fraction of high-interest-rate loans originated by the bank, and noincome measures the fraction of loans without income documentation originated by the bank in 2006. Adjusted R -squared and number of observations are provided in the bottom rows. All standard errors are clustered at the bank level.

5.11 Table 10: the role of demand deposits

Read:

- The coefficient on *after*preotd* remains positive at about 0.74 to 0.94.
- The triple-interaction *after*preotd*dd* is negative, between about -3.29 and -3.99, and statistically significant across specifications.
- The paper translates the estimate into about 0.24 percentage points lower defaults for a one-standard-deviation increase in the demand-deposit ratio, which is about 18% of the sample median default rate.

Economic message. High-OTD lending is most problematic at banks funded relatively more by non-demandable or wholesale sources. Demand deposits appear to discipline risk-taking rather than intensify it.

6 Short summary

The paper studies whether banks that relied heavily on originate-to-distribute lending before the subprime crisis made worse mortgages. Using bank call reports and HMDA data, it shows that banks with high pre-crisis OTD activity sold fewer mortgages after the 2007 market disruption and then suffered significantly larger mortgage chargeoffs and non-performing mortgage ratios. The result survives bank fixed effects, exclusion of large banks, matched-sample tests based on geography and borrower observables, matching on risky-loan spreads, and alternative mechanism checks based on unsold “stuck” loans. The paper also shows that high-OTD banks charged less differentiated spreads to risky borrowers and that the OTD effect is strongest among low-capital banks and banks with less reliance on demand deposits. The overall conclusion is that securitization weakened screening incentives in economically important ways.

Table 10
The effect of demand deposits

Dependent Var:	All Banks				Excludes Large Banks			
	Model 1 NPA		Model 2 NPA		Model 3 NPA		Model 4 NPA	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
<i>after</i>	0.5045	(3.66)	-0.2957	(-1.38)	0.4479	(3.00)	-0.3404	(-1.52)
<i>after*dd</i>	-1.6465	(-2.43)	-1.6697	(-2.44)	-1.3884	(-1.98)	-1.4100	(-2.02)
<i>after*preotd*dd</i>	-3.9949	(-2.34)	-3.7627	(-2.25)	-3.4250	(-2.09)	-3.2896	(-2.04)
<i>after*preotd</i>	0.9364	(2.79)	0.8260	(2.58)	0.8294	(2.52)	0.7423	(2.32)
<i>after*premortgage</i>	0.5526	(1.04)	0.7312	(1.43)	0.3507	(0.68)	0.5570	(1.07)
<i>logta</i>	0.2222	(0.51)	0.3085	(0.74)	0.6675	(1.39)	0.6942	(1.42)
<i>cil/ta</i>	2.7117	(1.44)	3.0167	(1.57)	2.4083	(1.21)	2.7033	(1.34)
<i>liquid</i>	1.3462	(0.92)	1.3215	(0.97)	-0.0527	(-0.05)	-0.0373	(-0.03)
<i>absgap</i>	-3.2555	(-3.91)	-3.0656	(-3.78)	-3.1305	(-3.70)	-3.0252	(-3.60)
<i>after*li</i>			0.2548	(2.53)			0.2648	(2.51)
<i>after*highrate</i>			1.9479	(4.07)			1.7502	(3.89)
<i>after*noincome</i>			0.6255	(2.33)			0.5534	(1.83)
R^2	0.7322		0.7358		0.7154		0.7173	
N	5397		5327		4977		4907	

This table provides the regression results of the following fixed-effect model:

$$\begin{aligned}
default_{it} = & \mu_i + \beta_1 after_t + \beta_2 after_t * preotd_t + \beta_2 after_t * dd_t \\
& + \beta_3 after_t * preotd_t * dd_t + \sum_{k=1}^K \beta_k X + \epsilon_{it}.
\end{aligned}$$

The dependent variable, $default_{it}$, is measured as the ratio of non-performing mortgages to the outstanding mortgage loans of bank i during quarter t . $after_t$ is a dummy variable that is set to zero for quarters before and including 2007Q1, and one after that. $preotd_t$ is the average value of OTD mortgages to total mortgages during quarters 2006Q3, 2006Q4, and 2007Q1; dd_t is bank i 's average demand deposits to total deposits ratio during quarters 2006Q3, 2006Q4, and 2007Q1; μ_i denotes bank fixed effects; X stands for a set of control variables. $premortgage$ is the average ratio of mortgage assets to total assets for 2006Q3, 2006Q4, and 2007Q1. $logta$ measures the log of total assets; cil/ta is the ratio of commercial and industrial loans to total assets; $liquid$ is the bank's liquid assets to total assets ratio; $absgap$ is the absolute value of one-year maturity gap as a fraction of total assets. li measures the average loan-to-income ratio of all loans issued by the bank in 2006. $highrate$ measures the fraction of high-interest-rate loans originated by the bank, and $noincome$ measures the fraction of loans without income documentation originated by the bank in 2006. Adjusted R -squared and number of observations are provided in the bottom rows. All standard errors are clustered at the bank level.

7 Conclusion

7.1 Core conclusion

The paper concludes that the originate-to-distribute model materially contributed to the poor quality of mortgages originated before the subprime crisis. Banks that used the model most aggressively were the ones whose mortgage portfolios deteriorated the most once market liquidity disappeared.

7.2 Economic intuition

The core intuition is simple. Screening soft information is costly. If the bank expects to keep the loan, spending those resources is worthwhile because the bank bears future losses. If the bank expects to sell the loan quickly, the private return to screening falls. That distortion is especially dangerous when markets later freeze and the bank ends up holding the very loans it underwrote too loosely.

7.3 Why the supporting evidence matters

The paper does more than document a correlation. It shows that high-OTD banks were less able to sell loans after the crisis, that worse post-crisis performance is concentrated in banks stuck with those loans, that the result survives careful matching on borrower observables and prices, and that loan-pricing dispersion is lower where screening appears weaker. Those pieces fit together into one coherent story.

7.4 What the heterogeneity says about incentives

The fact that the effect is strongest for low-capital banks and for banks with lower demand-deposit funding is especially informative. It suggests that the same securitization technology can be used very differently depending on a bank's balance-sheet discipline and funding model.

7.5 Policy takeaway

The paper implies that regulators should think about securitization and underwriting together. Capital rules, funding structure, retention incentives, and screening incentives are linked. The quality of mortgage-backed securities depends in part on the identity and incentives of the originating lender.

7.6 Bottom line

The big lesson is that the mortgage crisis was not only about bad aggregate shocks or falling house prices. It was also about incentive problems inside the lending chain. The OTD model made those problems worse by weakening the connection between loan origination and loan ownership.